

Two-Stage Frequency Acquisition for Over-the-Air LTE Cell Search and SIB1 Recovery

A Band 28 case study with an ADALM-Pluto SDR,
cross-validated with srsRAN 4G

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Professional context	Enextgen Wireless Limited · 2022
Original project	2022 · capture 25 August 2022
Retrospective validation	October 2026 · MATLAB rerun and srsRAN 4G 25.10
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4 October 2026 · Oluwanifemi Adelana Ogunjemilua

Document control

This report documents a 2022 project and a retrospective validation of its archived capture carried out in 2026. The 2022 work is the over-the-air capture and the MATLAB receiver in the [4G-LTE-Cell-Search-MIB-and-SIB1-recovery](#) repository; the 2026 work is a rerun of that receiver on the archived capture, an independent decode of the same capture with srsRAN, and the analysis built on them.

Item	Detail
Author	Oluwanifemi Adelana Ogunjemilua
Professional context	Enextgen Wireless Limited, 2022
Original project	2022: over-the-air capture on 25 Aug 2022; 4G-LTE-Cell-Search-MIB-and-SIB1-recovery repository published 21–22 Sep 2022
Retrospective validation	October 2026: original MATLAB receiver rerun on the archived capture; independent decode with srsRAN 4G 25.10 (release tag <code>release_25_10</code> , commit <code>6bcbd9e</code>)
Repository	https://github.com/Oluwanifemilua/4G-LTE-Cell-Search-MIB-and-SIB1-recovery
Report type	Technical report on the 2022 project, with 2026 retrospective validation
Report date	October 2026; decoded values from the MATLAB rerun, cross-checked with srsRAN
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Executive summary

This project used an ADALM-Pluto SDR to capture and decode a live Band 28 LTE downlink. I recorded a 40 ms over-the-air capture at 773 MHz and processed it with the MATLAB LTE Toolbox receiver, after adding a coarse frequency-offset search to handle the Pluto's oscillator error.

Rerunning that receiver on the archived capture in October 2026 recovered a decodable LTE cell: PCI 63, 10 MHz bandwidth, FDD operation and two antenna ports. The MIB was decoded at SFN 425, and SIB1 passed its CRC in two separate transmissions at SFN 426 and 428.

The decoded SIB1 identified PLMN 621-30 (MTN Nigeria), TAC 0x5828 and LTE Band 28. The coarse search selected +15 kHz and the fine estimator a further –1.79 kHz, a total carrier-frequency offset of about +13.2 kHz (17.1 ppm). That is outside the ± 7.5 kHz range of the receiver's original CP-based correction. The oscillator

uncertainty of a low-cost SDR can exceed the fine estimator's unambiguous range; the coarse PSS-based search ahead of cell search is what made the cell recoverable.

The same capture was also decoded independently with srsRAN 4G 25.10, a separately implemented open-source LTE receiver. It reproduced the cell identity, MIB parameters and both SIB1 transport blocks. The SIB1 bytes from the two receivers are identical, and their ASN.1 decode gives the field values above.

Background and objectives

The objective, as stated in the [4G-LTE-Cell-Search-MIB-and-SIB1-recovery](#) repository, was to capture a live over-the-air (OTA) 4G LTE signal with an ADALM-Pluto SDR and demodulate it in MATLAB to obtain the MIB and SIB1. This acquisition process is required before an LTE UE can proceed with network access.

A receiver that completes this chain on a commercial carrier shows that the capture front end, timing and frequency synchronisation, and the downlink control and shared-channel decoders all work together. The same acquisition chain is relevant to network scanners, drive-test tools and SDR-based LTE analysis.

Technical question. Can a low-cost SDR with a ± 25 ppm frequency reference capture a commercial LTE carrier well enough for a standard receiver chain to recover the cell identity, MIB and SIB1, and what must be added to the stock receiver to make that possible?

Contributions. The work makes the following engineering contributions:

- A two-stage frequency acquisition for the LTE Toolbox receiver: a coarse PSS-correlation search ahead of the stock CP-based estimator, extending the acquirable offset beyond ± 7.5 kHz.
- An end-to-end over-the-air demonstration on a commercial Band 28 carrier: cell search, MIB and SIB1 recovered from one archived 40 ms capture.
- A retrospective cross-check of the archived data: the original receiver rerun unchanged, a separately implemented srsRAN decode with negative controls, and byte-level agreement of the recovered SIB1.
- An engineering analysis that ties the decoded parameters to the LTE specifications: frequency-offset budget, transport-block derivation, decoding margin and system-information schedule.

Scope of the work:

- **In scope:** one OTA capture of one FDD carrier in E-UTRA Band 28; cell search; MIB decoding; SIB1 recovery through PCFICH, PDCCH and PDSCH.
- **Out of scope:** SIB2 and later SIBs, connected-mode procedures, multi-cell scanning, and repeated or statistical measurement campaigns.

Technical background: LTE downlink acquisition

A UE acquires an LTE cell in a fixed order, and each step unlocks the next. Synchronisation signals give timing and identity, the PBCH gives bandwidth and frame number, and only then can the control region be read to find SIB1 (3GPP TS 36.211, 36.212, 36.213, 36.331).

Step	Signal or channel	Where it sits (FDD, normal CP)	What it yields
1	PSS: length-63 Zadoff-Chu sequence, roots 25, 29, 34	Last OFDM symbol of slots 0 and 10, central 62 subcarriers	$N_{ID}(2)$ in $\{0, 1, 2\}$; 5 ms symbol timing
2	SSS: two interleaved length-31 m-sequences	Symbol just before the PSS	$N_{ID}(1)$ in $\{0 \dots 167\}$; 10 ms frame timing; CP length and duplex mode from the PSS-SSS spacing
3	Cell-specific reference signals	Fixed pattern set by the PCI	Channel and noise estimates for coherent demodulation
4	PBCH: QPSK, tail-biting convolutional code, 40 ms TTI	First 4 symbols of slot 1 in subframe 0, central 72 subcarriers	24-bit MIB: downlink bandwidth, PHICH configuration, 8 MSBs of the SFN; antenna-port count from the CRC mask; 2 SFN LSBs from the TTI phase
5	PCFICH	Symbol 0 of every subframe, 16 REs	CFI: number of control symbols (1–3)
6	PDCCH, common search space	First CFI symbols	DCI format 1A or 1C with CRC masked by SI-RNTI (0xFFFF)
7	PDSCH / DL-SCH: QPSK, turbo code, 24-bit CRC	Resource blocks given by the DCI, subframe 5 of even frames	SIB1 transport block, sent every 80 ms with four redundancy versions

The physical cell identity combines the two synchronisation results:

$$PCI = 3 \times N_{ID}(1) + N_{ID}(2), \text{ giving 504 identities, 0 to 503}$$

SIB1 carries the PLMN identity, tracking area code, 28-bit E-UTRAN cell identity, cell-selection threshold, frequency band indicator, and the schedule for all other SIBs. Decoding it shows the receiver can read live scheduled traffic, not only the fixed-format broadcast channel.

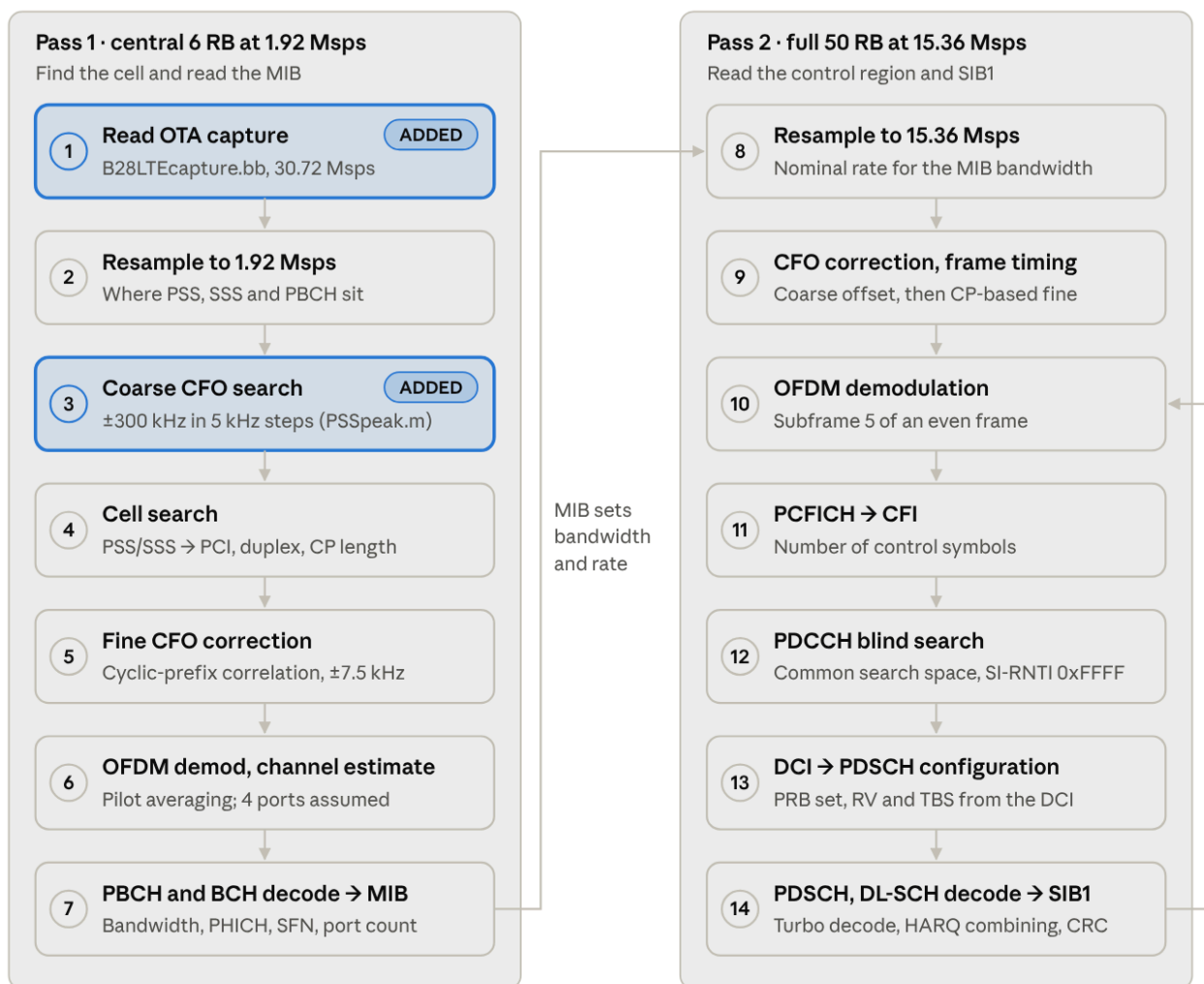
System design and implementation

The receiver is based on the MathWorks LTE Toolbox example *Cell Search, MIB and SIB1 Recovery*. I modified it to read the Pluto capture and added a coarse frequency-offset search before cell search.

The MIB from the central 6 RB unlocks the full-band SIB1 decode

☒ Added in this project (2022), also tagged ADDED

☐ From the MathWorks LTE Toolbox example, unchanged



Receiver chain in *SIB1RecoveryExample_wCoarseFr.m* · 14 numbered steps in 2 passes; the coarse CFO search (step 3) tests 121 offsets

The receiver runs in two passes. The first works on the central 6 resource blocks, because PSS, SSS and PBCH sit there whatever the carrier bandwidth. The MIB it decodes gives the bandwidth, which sets the sample rate for the second pass.

What was changed. By inspection against the published example, the main script differs in two blocks:

1. **I/Q source (step 1).** The stock `load eNodeB0output.mat` lines are commented out. A `comm.BasebandFileReader` reads `B28LTEcapture.bb` and takes the sample rate from the file.
2. **Coarse CFO search (step 3).** The stock receiver corrects frequency offset only by cyclic-prefix correlation, which is limited to ± 7.5 kHz (half the subcarrier spacing). The added loop shifts the 1.92 Msps waveform across ± 300 kHz in 5 kHz steps, scores each step with `PSSpeak.m` (best PSS correlation over $N_{ID}(2) = 0, 1, 2$), and applies the winner before cell search. The same coarse value is reapplied in pass 2.

The remaining stages use the toolbox reference functions and helpers unchanged.

Who did what.

Work	Source
Over-the-air Band 28 capture with the Pluto	Project work, 2022
Pluto I/Q input via <code>comm.BasebandFileReader</code>	Project work, 2022
Coarse CFO search and the <code>PSSpeak.m</code> metric	Project work, 2022
Core decoding chain: cell search, PBCH, PCFICH, PDCCH, PDSCH, DL-SCH	MathWorks LTE Toolbox example
Helper functions and plots	MathWorks LTE Toolbox example
Independent decode with srsRAN and Python sync, and the engineering analysis	Retrospective validation, 2026
Rerun of the 2022 MATLAB receiver on the archived capture	Retrospective validation, 2026

File	Size	Origin	Role
SIB1RecoveryExample_wCoars eFr.m	27.7 KB	MathWorks example (© 2010–2019), modified	Main receiver script
PSSpeak.m	0.4 KB	Added in this project (no MathWorks header)	PSS peak metric for the coarse CFO search
hPDSCHConfiguration.m	3.0 KB	MathWorks helper (© 2010–2016)	DCI to PDSCH configuration and transport block size
hSIB1RecoveryExamplePlots. m , hPlotPositions.m	2.9 KB, 1.5 KB	MathWorks helpers	Spectrum, correlation, constellation and channel plots
B28LTEcapture.bb	19.7 MB	OTA capture, 25 Aug 2022	Input I/Q data (see Experimental setup)
eNodeB0output.mat	415 KB	MathWorks sample capture, file dated 8 Aug 2013	Stock data; its load lines are commented out, so it is unused
CellSearch_MIB_SIB_PSS_SSS Correlation.png	60 KB	Figure the stock example embeds	Discussed under Limitations
README.md	0.2 KB	Project	One-sentence objective

Experimental setup and method

One 40 ms over-the-air capture was taken on 25 Aug 2022 with an ADALM-Pluto tuned to 773 MHz, inside the Band 28 downlink (758–803 MHz). Every parameter below comes from the metadata header that MATLAB wrote into the [capture file](#), `B28LTEcapture.bb`.

Parameter	Value recorded in the capture file
Receiver	ADALM-Pluto (AD9361), MATLAB <code>SDRReceiverConfiguration</code> ; RadioID <code>usb:0</code> , uri <code>ip:10.0.0.200</code>
Centre frequency	773.000 MHz
Sample rate	30.72 Msps
Length	1,228,800 complex samples = 40 ms = 4 radio frames, one contiguous burst
Gain control	AGC, slow attack
Front-end corrections	Quadrature, RF DC and baseband DC correction on
Analog filter	18 MHz RF bandwidth, custom filter off
Channels	1 receive channel
Sample format	Complex double, values on a 1/2048 grid (12-bit ADC scale)
Timestamp	25-Aug-2022 12:16:25 (time zone not recorded)

Not recorded anywhere in the 4G-LTE-Cell-Search-MIB-and-SIB1-recovery repository: antenna, location, cable or attenuator, and the console output of the original 2022 run. No attempt is made to reconstruct these unrecorded conditions; conclusions are restricted to quantities supported by the archived capture, the receiver output and the subsequent validation.

Capture length against the decoding need. The script's own guidance asks for at least 11 subframes to catch one PBCH block and 21 to catch one SIB1 transmission. The 40 subframes captured cover both, and hold two SIB1 transmissions. They fall short of the 81 subframes needed to collect all four SIB1 redundancy versions, so only RV 2 and RV 3 were available. srsRAN decoded each transmission on its own; MATLAB soft-combined the SFN 428 transmission with SFN 426.

Processing. The 2022 MATLAB receiver was rerun unchanged on the archived capture in October 2026; its console output and figures appear in Results. As an independent check, the same capture was decoded with srsRAN 4G 25.10. Both receivers recovered the same cell identity, MIB parameters and SIB1 transport blocks.

Results

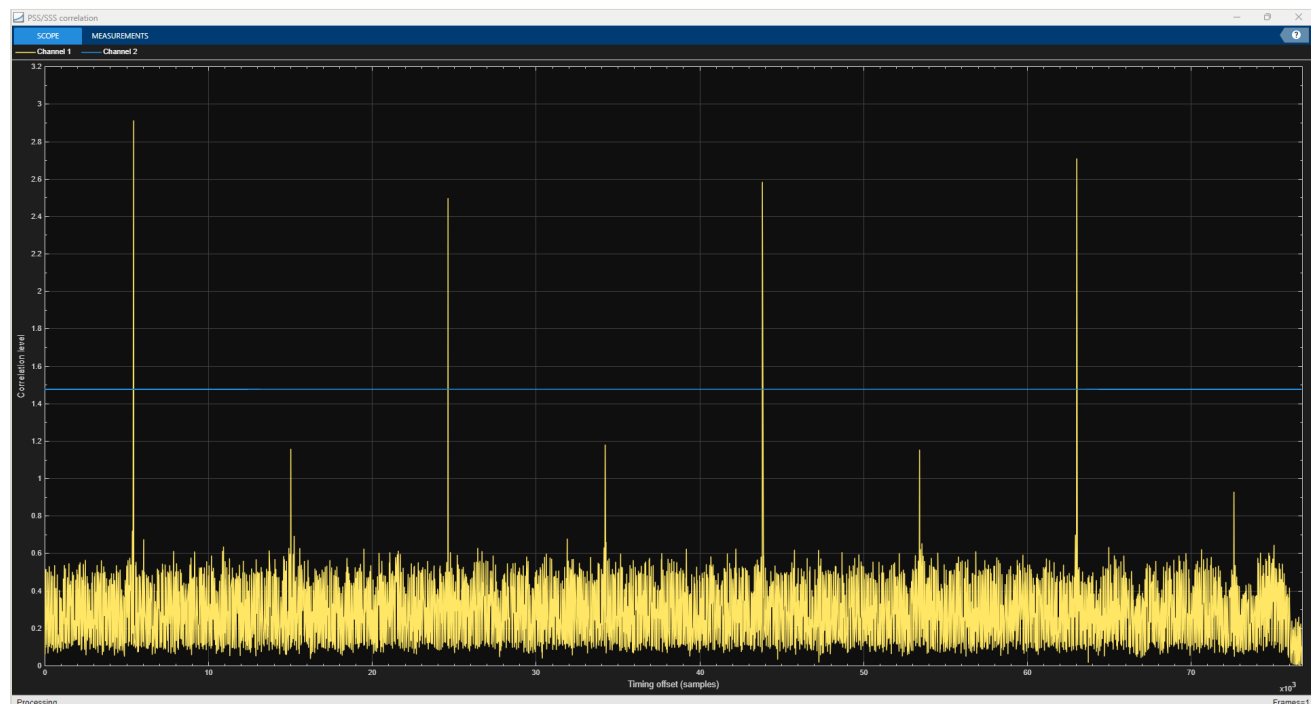
The MATLAB rerun and the srsRAN decode of the archived capture agree on a complete, decodable cell: PCI 63 on a 10 MHz FDD carrier, the MIB at SFN 425, and SIB1 with a CRC pass in both even frames (SFN 426 and 428). SIB1 identifies PLMN 621-30 in Band 28.

MATLAB receiver output

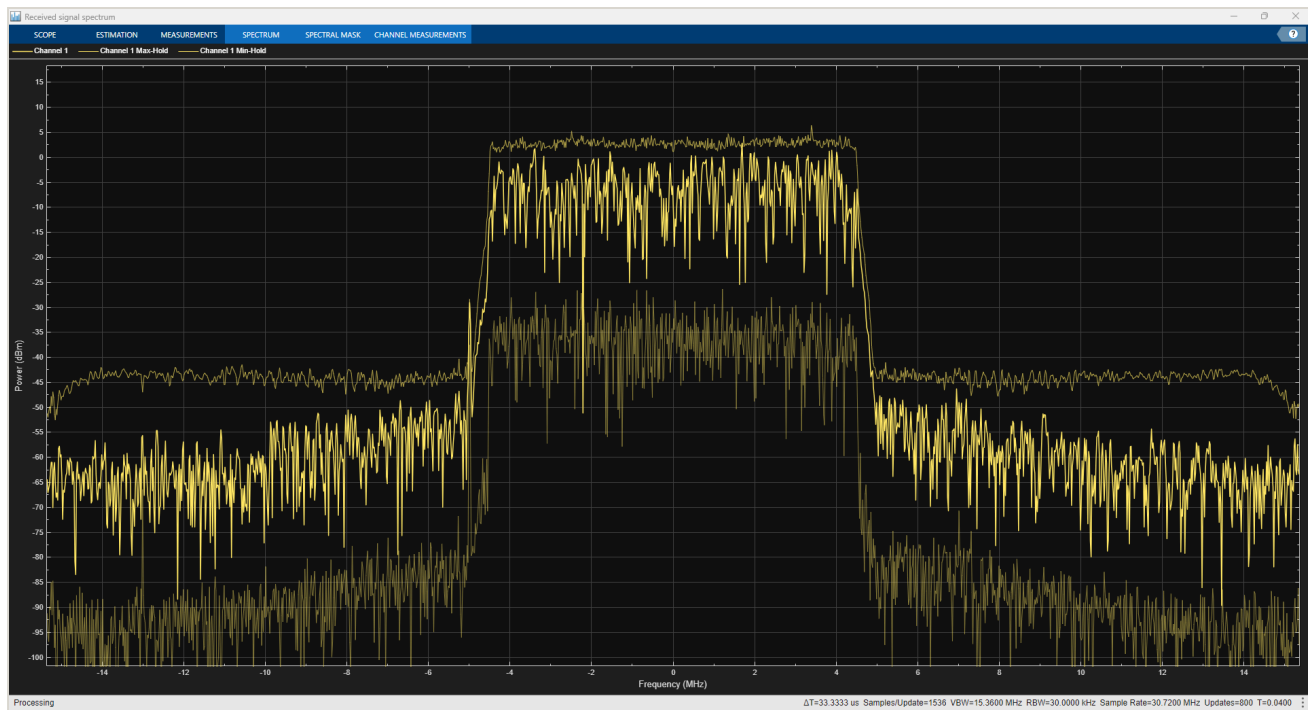
The 2022 receiver, `SIB1RecoveryExample_wCoarseFr.m`, was rerun unchanged on `B28LTEcapture.bb` in October 2026. Its console reported:

Step	MATLAB console output
3. Coarse CFO search	<code>coarseFreqOffset = 15000.0</code> (Hz)
4. Cell search	NCellID 63, FDD, normal CP; frame start 5,438 samples into the capture at 1.92 Msps
5. Fine CFO correction	-1,788 Hz in pass 1; -1,796 Hz in pass 2
7. MIB	NDLRB 50, CellRefP 2, PHICH normal, Ng = One, NFrame 425
9. Frame timing, pass 2	43,503 samples at 15.36 Msps
SFN 425	Skipped: odd frame, no SIB1
SFN 426, SIB1	CFI 2; DCI 1A, MCS 2, RV 2, TPC 1; 6 PRB, QPSK, transmit diversity; PDSCH RMS EVM 53.7%; CRC 0
SFN 428, SIB1	CFI 2; DCI 1A, MCS 2, RV 3, TPC 1; soft-combined with SFN 426; PDSCH RMS EVM 49.6%; CRC 0

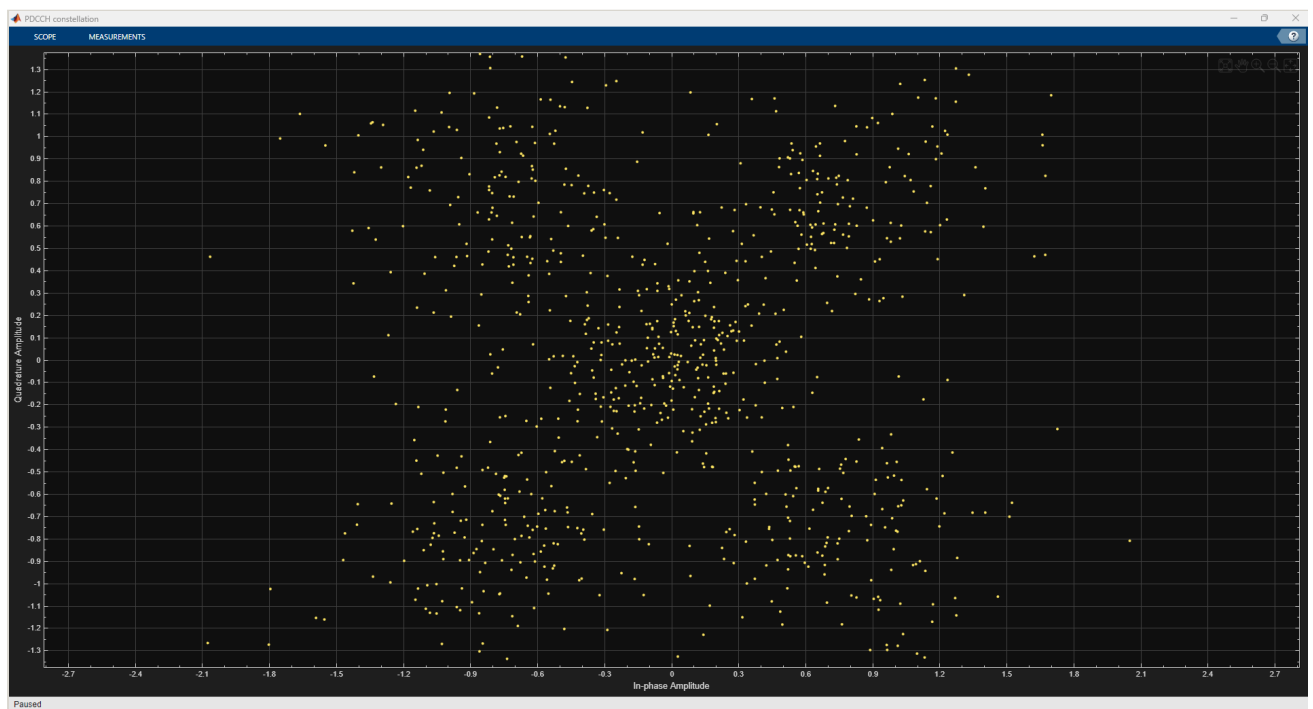
The PDSCH EVM of 53.7% and 49.6% corresponds to an SNR of about 5.4 and 6.1 dB, consistent with the SSS-based estimate under Engineering analysis.



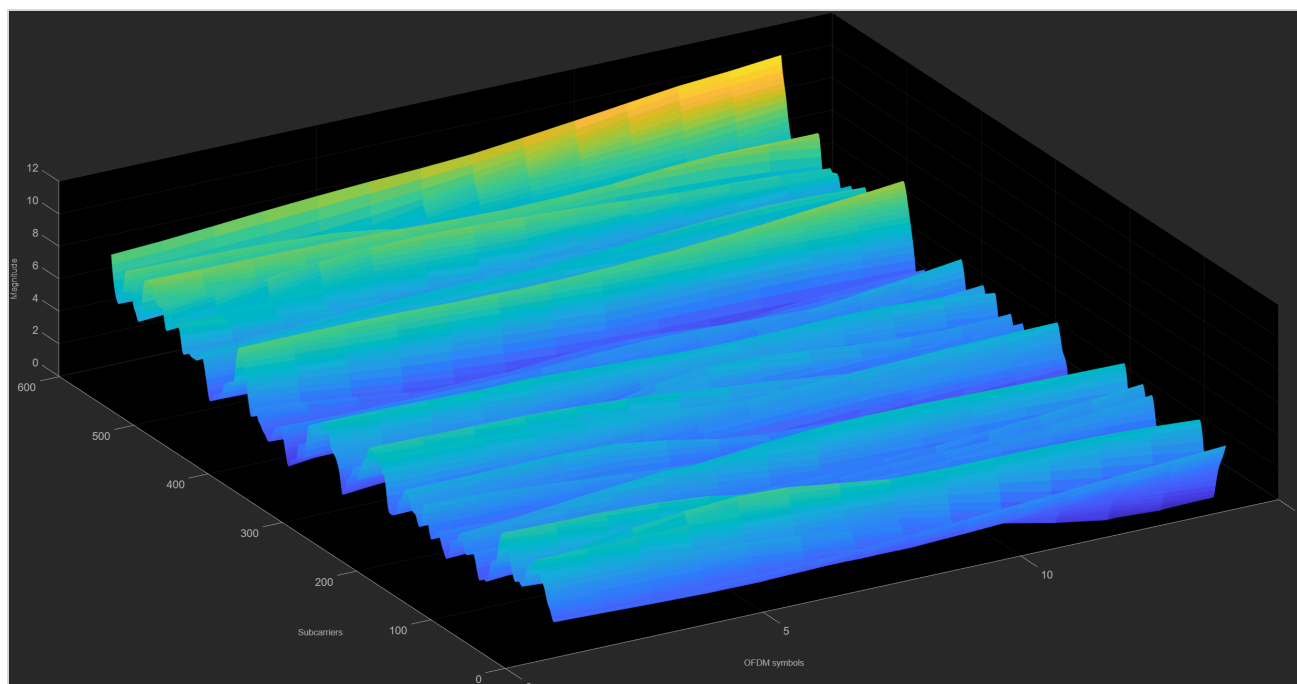
PSS/SSS correlation from the rerun, 1.92 Msps. The tall peaks, 19,200 samples (10 ms) apart, are subframe 0, where PSS and SSS both match; the smaller peaks between them are subframe 5. The blue line is the weak-correlation threshold. This is the Band 28 counterpart of the stock figure in the repository.



Received spectrum at 30.72 Msps: about 9 MHz occupied, consistent with the 50 PRB the MIB reports. Levels are uncalibrated (AGC on).



PDCCH symbols in a SIB1 subframe: four QPSK clusters spread by noise, and a cluster at the origin, consistent with unused control-channel elements sent at zero power.

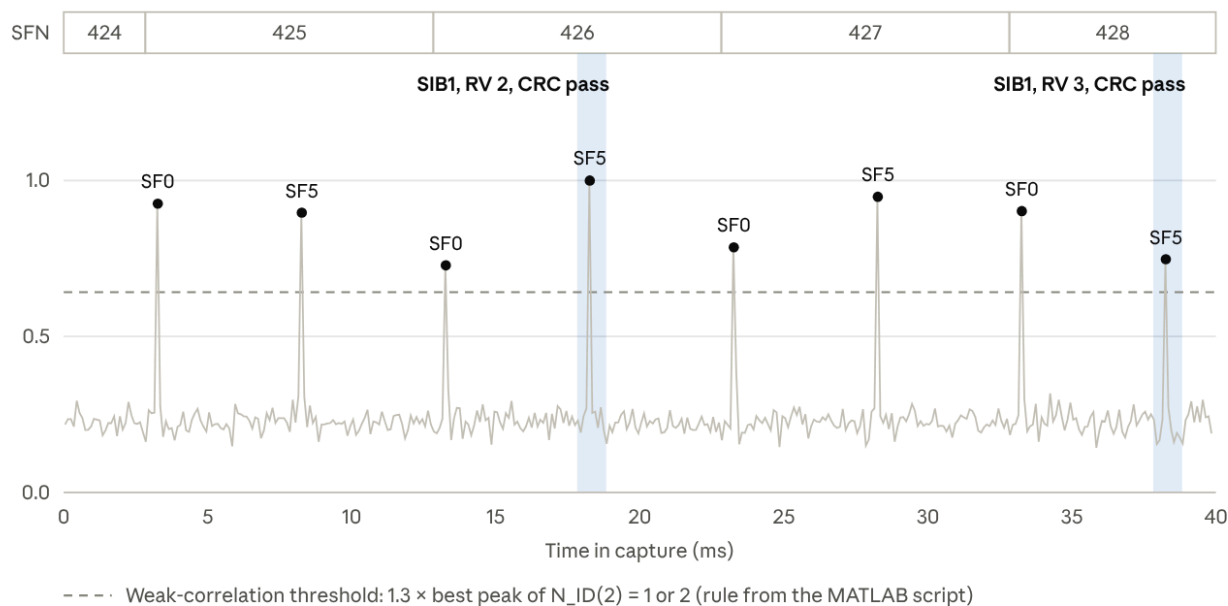


Channel magnitude estimate over 600 subcarriers and 14 OFDM symbols of a SIB1 subframe. The ripple across frequency shows frequency-selective multipath; the channel barely changes within the subframe.

Independent decode of the same capture

PSS peaks every 5 ms lock PCI 63; SIB1 decodes in both even frames

Band 28 capture at 1.92 Msps, PSS correlation for $N_{ID}(2) = 0$, normalised to its peak, 0.1 ms max-hold bins



B28LTEcapture.bb (25 Aug 2022), 2026 validation decode · 40 ms, 399 bins, 8 PSS peaks

The eight PSS peaks sit 9,600 samples (5.000 ms) apart. SSS gives $N_{ID}(1) = 21$ at seven of the eight, alternating subframe 0 and 5, so $\text{PCI} = 3 \times 21 + 0 = 63$. The last SSS did not resolve cleanly; its subframe label follows from the 5 ms spacing. Every peak clears the script's weak-correlation threshold: 0.73 to 1.00 against 0.64.

Cell identity and MIB

Quantity	Value	How it was obtained
Physical cell ID	63 ($N_{ID}(1) = 21$, $N_{ID}(2) = 0$)	MATLAB cell search and Python PSS/SSS search agree; PBCH fails when PCI 64 is forced
Duplex, cyclic prefix	FDD, normal	PSS–SSS spacing; Band 28 is an FDD band
Carrier frequency offset	+13.2 kHz (17.1 ppm at 773 MHz)	MATLAB: 15 kHz coarse plus –1.79 kHz fine. The Python PSS sweep, on a 100 Hz grid, gave +13.7 kHz
Frame timing	Frame boundary 2.832 ms into the capture	MATLAB and Python both find 5,438 samples at 1.92 Msps
Bandwidth	50 PRB (10 MHz)	MIB <code>dl-Bandwidth</code> ; occupied spectrum spans about ± 4.5 MHz
PHICH	Normal duration, $N_g = 1$	MIB
System frame number	425 for the first full frame	MATLAB NFrame 425; srsRAN MIB value 424 plus PBCH block offset 1
Antenna ports	2	MATLAB CellRefP 2; srsRAN's PBCH CRC passes only with the 2-port mask
PBCH SNR estimate	about 10 dB	srsRAN reference-signal estimate

SIB1 decoding chain

Stage	SFN 426, subframe 5	SFN 428, subframe 5
CFI	2	2
DCI	Format 1A, SI-RNTI, CCE 0, aggregation level 2	Same
Allocation	RIV 250, 6 PRB	Same
Modulation, MCS	QPSK, MCS 2	Same
Transport block	144 bits	144 bits
Redundancy version	2	3
Turbo decoder	Converged in 2 of 10 iterations	Converged in 2 of 10 iterations
CRC	Pass	Pass
Combining	First transmission, decoded alone	srsRAN: alone; MATLAB: soft-combined with SFN 426
PDSCH RMS EVM (MATLAB)	53.7%	49.6%

The redundancy versions match the SIB1 schedule in TS 36.321: $RV = \lceil 3k/2 \rceil \bmod 4$ with $k = (SFN/2) \bmod 4$, which gives 2 for SFN 426 and 3 for SFN 428. srsRAN decoded each transmission alone; MATLAB combined the second with the first. Both passed. The code rate is low, $(144 + 24 \text{ CRC bits}) / 1,584 \text{ coded bits} = 0.106$, which explains clean decoding at a modest SNR. Both transmissions carry the same 18 bytes: `40 58 84 61 58 28 56 5a e0 a8 21 b0 90 81 04 59 60 00`.

SIB1 contents

The MATLAB script checks the SIB1 CRC but does not parse the message. The SIB1 bits it recovered (`sib1{1}` , SFN 428) are byte-identical to srsRAN's, `405884615828565ae0a821b0908104596000` , and the field values below are the ASN.1 decode of those bytes.

Field	Decoded value	Interpretation
PLMN identity	MCC 621, MNC 30	Nigeria; 621-30 is MTN Nigeria (mcc-mnc.org)
Tracking area code	0x5828	22,568
Cell identity	0x565AE0A	eNB ID 353,710, local cell 10
Access	Not barred; intra-frequency reselection allowed; not a CSG cell	Open commercial cell
q-RxLevMin	-62	-124 dBm minimum RSRP for cell selection
Frequency band indicator	28	Agrees with the 773 MHz tuning
SI scheduling	SI-1 every 16 frames (SIB2, SIB3); SI-2 every 32 frames (SIB5)	SIB2 always rides in the first SI message; no SIB4 is scheduled
SI window, value tag	40 ms; 11	—

Independent validation (2026)

Three kinds of evidence are kept distinct in this report. **Verification** checks the decoded values against the LTE specifications (TS 36.211 to 36.331), as in the consistency checks below and the Engineering analysis.

Independent validation decodes the same capture with a separately implemented receiver, srsRAN 4G.

Reproducibility rests on the archived capture and code: the 2022 MATLAB receiver was rerun unchanged in 2026, and the validation scripts regenerate the srsRAN results.

For independent validation, the 2022 I/Q capture was processed with srsRAN 4G 25.10, built from the release tag `release_25_10` (commit `6bcbd9e` , released 26 Jan 2026). The same decode was also run on the later `master` commit `bef8680` (10 Sep 2026); its output was byte-identical. srsRAN existed in 2022, having been renamed from srsLTE in release 21.04, but it was not part of the original workflow.

The two receivers agree on every parameter both report:

Parameter	MATLAB, 2022 receiver (rerun 2026)	srsRAN 4G 25.10	Agree
PCI	63	63	Yes
Bandwidth	NDLRB 50	50 PRB	Yes
Antenna ports	CellRefP 2	2	Yes
PHICH	Normal, Ng = One	Normal, Ng = 1	Yes
First full frame	NFrame 425	SFN 424 + block offset 1 = 425	Yes
Frame start at 15.36 Msps	43,503 samples	43,504 (Python sync)	Within 1 sample
CFI, SFN 426 and 428	2, 2	2, 2	Yes
SIB1 DCI	Format 1A, MCS 2, RV 2 then 3	Format 1A, MCS 2, RV 2 then 3	Yes
SIB1 allocation	6 PRB, QPSK	6 PRB, QPSK, 144-bit block	Yes
SIB1 CRC, SFN 426 and 428	Pass, pass	Pass, pass	Yes
SIB1 bytes	4058...6000 (sib1{1}, 18 bytes)	4058...6000 (18 bytes)	Identical

The validation also forced incorrect antenna-port and PCI settings. Each of these made the PBCH decode fail, as expected.

Step	Tool	Output
1. Read capture	<code>bbfile.py</code>	Parses the 2,105-byte header and 1,228,800 samples; checks sample count against the recorded 40 ms
2. Synchronise	<code>lte_sync.py</code> (PSS/SSS from TS 36.211 §6.11)	CFO about +13.7 kHz on a 100 Hz grid (refined by srsRAN's own tracking), PCI 63, frame boundary at 2.832 ms
3. Decode	srsRAN 4G 25.10 <code>pdsch_ue</code> , file mode	MIB, PCFICH, PDCCH blind search, PDSCH and DL-SCH with CRC checks
4. Interpret	<code>decode_sib1.py</code> (pycrate, TS 36.331)	SIB1 fields from the decoded bytes

srsRAN's file mode has no cell search, so step 2 supplies the cell ID, offset and timing. A five-line patch to `pdsch_ue` prints the decoded SIB1 bytes and changes no decoding logic.

Negative controls. These are a discriminating check: if the decode depends on the recovered cell parameters, forcing a wrong value must break it. Forced settings and the PBCH result:

Forced setting	Why it must fail	Result
1 antenna port	The PBCH CRC is masked by the port count	No MIB
4 antenna ports	Same mask, different value	No MIB
PCI 64	PBCH scrambling and reference-signal positions follow the PCI	No MIB
PCI 63, 2 ports (true)	—	MIB at SFN 425

Consistency checks. The decoded values were checked against the LTE configuration and against each other:

- **Bandwidth:** the MIB's 50 PRB is read from the payload, not from the decoder settings, and matches the ± 4.5 MHz occupied spectrum.
- **Redundancy versions:** the DCI's RV 2 and RV 3 match the TS 36.321 schedule for SFN 426 and 428.
- **Band:** SIB1 reports band 28, as expected at 773 MHz.
- **Repeatability:** two separate SIB1 transmissions decode to identical bytes.
- **Syntax:** the 144-bit block decodes as a valid 131-bit SIB1, and the remaining 13 bits are all zero, as transport-block padding should be.

Reproducibility and provenance

Each result traces to a fixed artefact:

- **Capture:** B28LTEcapture.bb, recorded 25 Aug 2022, with the metadata header MATLAB wrote into it.
- **Receiver:** SIB1RecoveryExample_wCoarseFr.m and PSSpeak.m as published in the repository in September 2022, rerun unchanged in October 2026.
- **Validation:** the validation/ package (scripts, five-line patch and reference outputs) and srsRAN 4G at tag release_25_10, commit 6bcbd9e.

Reproducing it. The scripts, patch and reference outputs are packaged as a `validation/` folder for the [4G-LTE-Cell-Search-MIB-and-SIB1-recovery](#) repository. `./run_validation.sh` runs all four steps and the controls in a few seconds once `pdsch_ue` is built.

Engineering analysis

The decoded parameters were also checked against the LTE specifications and the Pluto frequency-accuracy specification.

Frequency-offset budget

The stock receiver estimates frequency offset from cyclic-prefix correlation, which measures phase over one useful symbol, $T_u = 1/15$ kHz = 66.7 μ s. Phase wraps at $\pm\pi$, which sets the unambiguous range:

$$\Delta f = \arg(\sum r[n] \cdot r^*[n + N]) / (2\pi \cdot T_u), \text{ unambiguous only for } |\Delta f| < 1 / (2 \cdot T_u) = 7.5 \text{ kHz}$$

Quantity	Value
Pluto reference accuracy	± 25 ppm (Analog Devices)
Worst-case offset at 773 MHz	± 19.3 kHz
Offset estimated from the capture	+13.2 kHz (17.1 ppm), from the MATLAB coarse and fine estimates
CP-only estimate	$13.2 - 15 = -1.8$ kHz, aliased: one whole subcarrier of error left uncorrected
Coarse search, 5 kHz grid	MATLAB selected +15 kHz; residual -1.79 kHz, inside the fine range
Largest residual a 5 kHz grid can leave	2.5 kHz, well inside ± 7.5 kHz

A one-subcarrier error shifts PSS, SSS and PBCH off their subcarriers, so without the coarse stage the cell cannot be acquired. The original implementation used a ± 300 kHz search range. This is considerably wider than the Pluto's ± 25 ppm specification requires. A ± 25 kHz range would give enough margin and cut the search from 121 to 11 points.

Resource allocation and transport block

The DCI's resource indication value decodes as six contiguous virtual resource blocks:

$$\text{RIV} = N_{\text{RB}} \times (L - 1) + \text{RB_start} \rightarrow 250 = 50 \times 5 + 0 \rightarrow L = 6 \text{ blocks, RB_start} = 0$$

- **Resource elements:** $6 \text{ PRB} \times 12 \text{ subcarriers} \times (14 - 2 \text{ control symbols}) = 864$. Reference signals for ports 0 and 1 take 4 REs per PRB in symbols 4, 7 and 11, which is 72. That leaves 792 REs, or 1,584 coded bits at QPSK, exactly what the decoder reports.
- **Transport block size:** with SI-RNTI and format 1A, $I_{\text{TBS}} = I_{\text{MCS}} = 2$, and the TPC field selects the N_{PRB} column, 2 or 3. TS 36.213 Table 7.1.7.2.1-1 gives 72 or 144 bits. The TPC field is 1 in the MATLAB output, which selects column 3: 144 bits, the size both receivers decoded.
- **Code rate:** $(144 + 24 \text{ CRC bits}) / 1,584 = 0.106$.

Information-theoretic decoding margin

For each channel, the spectral efficiency η (information bits per QPSK symbol) sets the minimum SNR an ideal code would need on an AWGN channel, $\text{SNR}_{\text{min}} = 2^{\eta} - 1$:

Channel	Info + CRC bits	Coded bits	Code rate	Bits per QPSK symbol	AWGN Shannon-limit SNR	Estimated SNR
PBCH, one 10 ms block	40	480	0.083	0.167	-9.1 dB	about 10 dB
SIB1 on PDSCH	168	1,584	0.106	0.212	-8.0 dB	5.4-6.1 dB (MATLAB EVM)

Both channels operated 13 to 19 dB above the AWGN Shannon-limit SNR for the corresponding spectral efficiency. This difference is not a direct coding margin: the bound assumes ideal coding on an AWGN channel and does not capture finite-blocklength, channel-estimation, synchronisation or implementation losses. It does show both channels well above the information-theoretic minimum, consistent with the MIB decoding from one PBCH block and srsRAN decoding each SIB1 transmission on its own, with the turbo decoder converging in 2 of 10 iterations.

The SNR itself was modest. srsRAN estimates about 10 dB on PBCH and PDCCH; SSS error after PSS-based equalisation gives 3.5 to 8.5 dB per 5 ms block; MATLAB's PDSCH EVM of 49.6-53.7% gives 5.4-6.1 dB. The out-of-band floor sits far below the in-band level, so the impairment is in-band. A weaker $N_{\text{ID}}(2) = 1$ correlation also appears, at 0.49 of the serving peak and aligned in time. A co-channel neighbour cell is one possible explanation, but this is tentative: SSS did not resolve a PCI for it, and no neighbour cell is claimed.

System information schedule

SIB1 maps SIB2 and SIB3 to SI message 1 every 16 frames (160 ms) and SIB5 to SI message 2 every 32 frames (320 ms), each in a 40 ms window. A capture is guaranteed to contain a full window when it is at least the period plus the window long: 200 ms for SIB2/SIB3 and 360 ms for SIB5. No SIB4 is scheduled, so the cell broadcasts no intra-

frequency neighbour list. Scheduling SIB5 shows that inter-frequency reselection is configured; the capture does not establish which carriers SIB5 lists, because its window was not captured.

Limitations

The main limitations are:

- **One capture.** A single 40 ms recording gives two SIB1 transmissions, with no repeatability or time-of-day data.
- **Archival note.** The original 2022 MATLAB console log was not retained. The receiver was rerun unchanged on the archived capture in October 2026, and its output is the one reported here; the source code and I/Q capture are archived in the [4G-LTE-Cell-Search-MIB-and-SIB1-recovery](#) repository. The 2026 srsRAN decode reproduces it independently.
- **Uncalibrated levels.** AGC was on, and the antenna, location and cabling were not recorded, so no absolute signal level or coverage can be stated.
- **Short capture.** 40 subframes cannot combine all four SIB1 redundancy versions or reach the SIB2, SIB3 and SIB5 windows.
- **Repository notes.** The [correlation PNG](#) is the stock example's reference image: its x-axis spans the 13,440 samples of `eNodeB0output.mat`, not the 76,800 of the Band 28 capture. The October 2026 rerun produced the Band 28 version, shown under Results. Some inline comments still describe the stock dataset.

Accordingly, the result is presented as a demonstrated case study and receiver-engineering validation, not as a statistical characterisation of LTE acquisition performance.

Conclusions

The project showed that an ADALM-Pluto can be used to capture and decode a live Band 28 LTE downlink. The archived 40 ms capture was enough to recover PCI 63, the MIB and two separate SIB1 transmissions.

The main receiver modification was the coarse CFO search. The captured carrier had an estimated +13.2 kHz offset at 773 MHz (15 kHz coarse, -1.79 kHz fine), outside the unambiguous range of the original CP-based correction. The coarse search brings the residual offset inside that range, which the rest of the LTE Toolbox chain needs in order to acquire the cell.

Rerunning the 2022 MATLAB receiver on the archived capture in October 2026 recovered the cell, MIB and both SIB1 transmissions with CRC pass. An independent srsRAN 4G 25.10 decode reproduced every parameter, and forced wrong settings failed as expected. The remaining limitations are the single short capture and the lack of calibrated RF-level measurements.

Future work

Three research directions follow from the limitations:

1. **Extended-duration system-information capture.** A capture of at least 81 ms would collect all four SIB1 redundancy versions. About 200 ms would guarantee a full SIB2/SIB3 window and about 360 ms the SIB5 window, which would show the inter-frequency carriers SIB5 lists.
2. **RF and frequency-reference calibration.** A GPS-disciplined reference and manual gain control would make CFO, absolute level and SINR measurements repeatable across captures, sites and times of day.
3. **Acquisition-search optimisation.** The coarse search used ± 300 kHz in 5 kHz steps, 121 points. A later receiver revision could derive the range from the oscillator envelope, for example ± 25 kHz (11 points), and test the speed-up against acquisition reliability.

Evidence register and sources

Timeline.

Artefact	Date and context
ADALM-Pluto over-the-air capture (<code>B28LTEcapture.bb</code>)	25 Aug 2022, original project
MATLAB receiver code (<code>SIB1RecoveryExample_wCoarseFr.m</code> , <code>PSSpeak.m</code>)	2022, original project
4G-LTE-Cell-Search-MIB-and-SIB1-recovery repository published (commit 5ee9b1f)	21–22 Sep 2022, original project
Decode of the archived capture with srsRAN 4G 25.10 (commit 6bcbd9e), Python sync and pycrate	October 2026, retrospective validation
Engineering analysis and this report	October 2026
Rerun of the 2022 MATLAB receiver on the archived capture: console log and four figures	October 2026, retrospective validation

The table below lists where each result comes from: a file in the [4G-LTE-Cell-Search-MIB-and-SIB1-recovery](#) repository or a step of the 2026 validation.

Result	Source artefact	Derivation
Capture date, 773 MHz, 30.72 Msps, 40 ms, AGC, Pluto	B28LTEcapture.bb header	Read directly from the 2,105-byte metadata header
Receiver design and the two modifications	SIB1RecoveryExample_wCoarseFr.m , PS Speak.m	Code reading; compared with the published MathWorks example
Stock data unused; PNG from stock data	eNodeB0output.mat , PNG	Commented-out load lines; sample-count comparison
PCI 63, CFO, frame timing	B28LTEcapture.bb	Python PSS/SSS search per TS 36.211
MIB, antenna ports, SFN	B28LTEcapture.bb	srsRAN pdsch_ue PBCH decode; 1-port, 4-port and PCI 64 controls fail
CFI, DCI, PDSCH grant, SIB1 bytes	B28LTEcapture.bb	srsRAN pdsch_ue with SI-RNTI; two CRC passes
SIB1 fields	SIB1 bytes, identical from MATLAB and srsRAN	pycrate UPER decode of BCCH-DL-SCH-Message
Forced-setting checks, consistency checks, reference outputs	validation/ folder	run_validation.sh; outputs in validation/expected/
CFO budget, TBS, decoding margin	Decoded values above	Derivation from TS 36.211, 36.213 and the Pluto specification
MATLAB cell search, CFO, MIB, CFI, DCI, SIB1 CRC and EVM; four figures	SIB1RecoveryExample_wCoarseFr.m on B28LTEcapture.bb	MATLAB console log and figures, October 2026 rerun

Sources:

- Project repository: <https://github.com/Oluwanifemilua/4G-LTE-Cell-Search-MIB-and-SIB1-recovery>
- [MathWorks: Cell Search, MIB and SIB1 Recovery](#) — the stock example; loads eNodeB0output.mat , CP-based CFO only (± 7.5 kHz)
- [srsRAN 4G 25.10 release](#) — tag release_25_10 , commit 6bcbd9e , published 26 Jan 2026; used for the 2026 retrospective validation, and cross-checked against master commit bef8680 (10 Sep 2026) with identical output. The project was renamed from srsLTE to srsRAN in release 21.04 ([release history](#))
- mcc-mnc.org: PLMN 621-30 — MTN Nigeria Communications Limited
- 3GPP TS 36.211 (physical channels), TS 36.212 (coding), TS 36.213 (physical layer procedures), TS 36.321 (MAC, SIB1 redundancy versions), TS 36.331 (RRC, SIB1 structure)
- [Analog Devices: ADALM-PLUTO phase noise and frequency accuracy](#) — ± 25 ppm initial accuracy